Decision Models on Bengali: Accuracy, Calibration and Robustness to Romanization

Ismail Hossain Polas
Jahangirnagar University
ismailhossainpolash@gmail.com

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Abstract

Decision models answer a typed question in one forward pass and return a probability for every option. I tested one hosted decision model, Jev, ten open decision models and one general-purpose language model on five Bengali tasks, zero-shot. Jev scored highest on all five tasks among the models run on the full test sets; on Belebele it reached 0.918 (95% CI 0.897 to 0.936). Rule-based romanization lowered every system's accuracy on every task it was applied to, by 13.5 to 35.9 points on average. To see how far tool-made romanization moves a system's answers compared with real typing, I used the Latin-script spelling that native speakers wrote for 1,500 Bengali Wikipedia sentences (the Dakshina dataset). These sentences have no labels, so I measured consistency with each system's own Bengali-script answer, not accuracy. All 12 systems kept that answer more often under the native-speaker romanization than under the rule-based one, by +3.3 to +12.9 points, and every interval excludes zero. Jev kept it 0.733 of the time against 0.672. A romanizer tuned on typed Banglish differed from the rule-based one by -1.8 to +4.5 points. The ordering of systems was almost the same under both spellings (Kendall $\tau = 0.89$). On average over three tasks, all 11 systems with full test sets had confidence above accuracy, though not on every task (3 of 32 model-task cells were the other way). One temperature fitted on a separate development set lowered calibration error in all 26 model-task cells where it was tried, for 9 systems, and it helped less on rule-based romanized text. Code, tables and per-sentence predictions are at `https://github.com/ihpolash/system-one-testing-for-bengali`.

1 Introduction

A decision model reads a text and a typed question and returns one probability per option, without generating anything. That makes the output easy to act on: a pipeline can accept confident answers and send the rest to a larger model or a person [4, 16]. Nearly all of the evidence for this comes from English. The one study that left English shows why that matters. On Chinese tasks Laya-multilingual scored 40.2% where Jev scored 68.4% [20]. For Bengali I found only the per-language Belebele and SIB-200 numbers that ship with the Jev benchmark [4]. I found no study that asked whether a decision model's confidence still means something in Bengali.

Bengali has a second difficulty. A great deal of everyday Bengali is typed in Latin letters, with no standard spelling. Work on other Indic languages and on Bangla dialects tests romanized text produced by a transliteration tool, and finds large losses [11, 12]. A tool applies one rule per letter. People do not. I found no measurement, for Bengali, of how far the loss on tool-made romanization differs from the loss on what native speakers actually write.

Study	Decision models	Bengali	Romanized	Calibration	Human-typed text
Deußer et al. [4]	Jev only	per-language file	Belebele, SIB-200	yes	no
Wang et al. [20]	trained model	no	no	yes	no
Khullar et al. [12]	no	no	tool-made	routing only	no
Jawad et al. [11]	no	dialects	tool-made	no	no
This paper	12 systems	5 tasks	rule-based, natural-style, native-speaker	yes	yes (1,500 sentences)

Table 1: What each closest study covers, from reading the papers. A "no" means the paper does not report it.

The Dakshina dataset makes that measurement possible: 10,000 Bengali Wikipedia sentences that native speakers romanized by hand [17]. It has no task labels, so I measure something else: how often a system's answer to a topic question survives the change of script (Section 4.2). Around it sits a zero-shot benchmark of twelve systems on five tasks, with calibration, risk–coverage and option-order analyses (Section 4). Every number here is produced by the released code, and the harness reproduces published numbers from independent groups for Jev, Laya-multilingual and laya-bangla (Section 4.5). This is not the first Bengali result for any decision model, since Jev's Belebele and SIB-200 numbers exist. Kev-9B, Kev-27B, d3-27B and Microsoft-Decision-1 are not covered.

2 Related work

Table 1 shows what the closest studies cover. In English, decision models have been checked for calibration [16], for annotation work with open models [9], for sensitivity to option names [19] and to option order [13], and for cost and wording robustness [18]. Other languages are thin. Chinese-Jev trains a Chinese model [20], and one paper compares Jev's cultural-value answers in English and Arabic [2]. Script robustness has been studied for generative models only [11, 12], and the Bengali half of the large romanization corpus of Gharami et al. [7] is generated by rule, not typed by people. For calibration and selective prediction I use the standard measures [6, 8]; the option-rotation test follows Zheng et al. [21]. The tasks are Belebele [3], SIB-200 [1], MASSIVE [5], SentNoB [10] and the Bengali translation of XNLI; the romanization data are from Dakshina [17].

3 Benchmark

Models. Jev (`jev-1.13.0`) is reached through its API. The open models ran on Kaggle GPUs: Laya-multilingual, an mmBERT-based encoder [15]; laya-bangla v2, a Bangla fine-tune of it (commit `4c5eb48932`); Julia-1, with 144M parameters; the vLLM-SR Decision 3.0 models d3-edge, d3-lite, d3-nano, d3-mini and d3-flash [14]; and Kev-0.8B and Kev-4B (release 1.0, tag `v1.0`), served with the project's own code in bf16 with its shipped temperature. Kev's model cards list English as its only language and name other languages as a known weakness. Kev-4B sits at the memory limit of a 15 GB T4, so for it I switched off CUDA-graph replay and the prefix cache, which change speed and memory but not the model, after a first run ran out of memory. Kev was scored on the test sets and on the Dakshina sentences only, so it is missing from the calibration and option-order results. d3-flash has 9B parameters and did not fit comfortably, so it ran 4-bit (NF4), on a hash-ordered subset of about 405 items per task and of 500 Dakshina sentences. Gemma-4-E4B-it is not a decision model; it serves as a baseline, with each option scored by the next-token probability of its label and the probabilities normalized over the options. Kev-9B and Kev-27B do not fit on that GPU, and d3-27B and Microsoft-Decision-1 (API only) were not run. Exact identifiers and

Dataset	Task	Options	Metric	Text origin	Test items (all / used)
Belebele	reading comprehension	4 options	accuracy	human translation (FLORES-200)	900 / 900
SIB-200	topic	7 options	accuracy	human translation (FLORES-200)	204 / 204
XNLI-bn	natural language inference	3 options	accuracy	machine translation of XNLI	4,895 / 1,500
MASSIVE bn-BD	intent	60 options	accuracy	human translation	2,974 / 1,500
SentNoB	sentiment	3 options	macro-F1	native social-media comments	1,586 / 1,586

Table 2: Tasks. XNLI-bn is the machine-translated Bengali XNLI, so its text is translationese; SentNoB is the only native Bengali set. MASSIVE is posed with all 60 intents; the test items use 59 of them. SentNoB's test split is that of a third-party mirror and its size differs between sources (Section 3, paragraph SentNoB split).

settings are in `docs/MODELS.md`.

Tasks. The five tasks are in Table 2. Each item is a single-choice question with English instructions and option names; only the passage or utterance changes between conditions. The exact question structure for every task is in `docs/PROMPTS.md` in the repository. I cut XNLI-bn and MASSIVE to hash-ordered subsets of 1,500 items so that any smaller run is nested in a larger one. Bengali option names changed Jev's accuracy by at most 0.8 points on SentNoB, XNLI-bn and SIB-200, so I kept English names throughout. The metric is accuracy, and macro-F1 for SentNoB.

SentNoB split. I used the test split of the Hugging Face mirror `sadi2oo3q/SentNoB`. Its train, validation and test splits hold 12,575, 1,567 and 1,586 comments, which add up to the 15,728 of the original paper. The BanglaBERT benchmark lists 1,567 test comments and the laya-bangla authors report 1,564, so the size of "the" test set differs between sources, and I could not find where the 19 extra comments against BanglaBERT come from. Every SentNoB number in this paper, including the released predictions, uses the same 1,586 comments. The mirror's test split overlaps its training split: 362 test comments (23%) also occur verbatim in its training split, and 27% occur in the training or validation split or earlier in the test split (exact match after whitespace normalization). This cannot help a zero-shot system, and laya-bangla's authors state that it was never trained on SentNoB; I cannot say that for the other systems. Dropping the overlapping comments leaves 1,161 and moves every system's macro-F1 by at most 2.1 points (Jev 0.623 to 0.605), with the order of systems nearly unchanged (Kendall $\tau = 0.96$; Table 12).

Romanization. *Rule-based romanization* maps each letter to Latin letters by a fixed rule set, without diacritics. It is applied to the test sets of four tasks (Belebele passages were not romanized). *Natural-style romanization* replaces each word with its most frequent spelling in a public corpus of typed Banglish (nahidstaq/bangla-transliteration-data) and falls back to rules tuned on held-out words from that corpus. That corpus turned out to contain 4,144 Dakshina sentences with their native-speaker romanization, so my first version had effectively seen the target spellings and scored 0.695 for Jev. I removed every pair that matches a Dakshina string, rebuilt the lookup from the remaining 34,870 sentences, re-checked the rule settings and reran; all natural-style numbers here use the rebuilt version. *Native-speaker romanization* is the Latin-script text in Dakshina, which native speakers wrote for each Bengali sentence without a transliteration tool; a second set of annotators transcribed it back to Bengali script to validate it [17].

Dakshina sentences. I kept the sentences of 6 to 40 words whose romanization has no Bengali script and dropped duplicates, which left 8,244 of the 10,000, and used the first 1,500 in hash order (500 for d3-flash, nested in them). There are no task labels. Every system answers one seven-way topic question about each sentence, with the SIB-200 topics, in four spellings: Bengali script, native-speaker, rule-based and natural-style romanization. The measure is agreement: the share of sentences whose top option is unchanged from the system's own answer on the Bengali-script sentence. I also report the mean total-variation distance between the two probability vectors, and the agreement expected by chance, $\sum_c p_{\text{Bengali}}(c)\, p_{\text{romanized}}(c)$ (0.191 on average). Agreement is not accuracy: a system that is wrong on the Bengali sentence can agree with itself on the romanization.

Statistics. Intervals are 95% percentile bootstraps (seed 0). On the test sets they resample premise or passage clusters (Belebele, XNLI-bn) or items, 1,000 times (400 for the SentNoB macro-F1, 500 for paired differences). On the Dakshina sentences they resample sentences 1,000 times, and every comparison between spellings is a paired difference on the same sentences. Calibration is measured with expected calibration error over 15 equal-width bins (ECE), the signed gap between confidence and accuracy, the Brier score and risk–coverage curves. The Brier score of an item is the sum over its K options of the squared difference between the predicted probability and the one-hot label, $\sum_{k=1}^{K}(p_k - y_k)^2$, and a task's score is the mean over items; it runs from 0 to 2, and a uniform guess scores $1 - 1/K$, which is 0.667 on the three-option tasks and 0.983 on MASSIVE. Scores from tasks with different K are therefore not comparable with each other; where I average them, it is an unweighted mean over XNLI-bn, MASSIVE and SentNoB (Table 13), and the per-task scores are in Table 11. ECE is computed within each task on the full test sets and then averaged without weights over XNLI-bn, MASSIVE and SentNoB (Table 7; per task in Table 10), because the tasks differ in the number of options and in label balance. Temperature scaling uses one scalar per model and task. It minimizes the mean negative log-likelihood over a grid of 41 log-spaced temperatures from 0.2 to 6, on a development split of 1,500 items (XNLI-bn, MASSIVE, SentNoB) that is disjoint from the test items and holds Bengali text only. The fitted temperature is then applied unchanged to the test items in Bengali and in rule-based romanization; no test label enters the fit. Belebele and SIB-200 have no development split, and d3-flash and Kev were not run on one, so they have no scaled results. The confidence gate uses the same development split: its threshold is the smallest one whose accepted dev accuracy reaches 90%, with at least 5% coverage. Option-order sensitivity uses three cyclic rotations of the options. I report no significance tests. Two sets of comparisons are inferential, each with a paired bootstrap interval: native-speaker against rule-based and against natural-style romanization on the Dakshina sentences, and the differences to Jev in Table 4. As a robustness check on the first, I repeated the native-speaker against rule-based interval at a Bonferroni-adjusted level for the 12 systems (99.6%, 5,000 draws): it still excludes zero for 11 of them (not for Julia-1, the system closest to its floor). Everything else (rankings by point estimate, calibration, the confidence gate, option-order sensitivity) is descriptive and uncorrected.

4 Results

4.1 Accuracy on Bengali script and rule-based romanization

Table 3 and Figure 1 give the main results. Every system is less accurate on rule-based romanization than on Bengali script on every task where both were run: by 13.5 to 35.9 points on average over the tasks, 18.1 for Jev. Jev has the highest point estimate on every task among the models run on the full sets, but the lead is not equally firm (Table 4, paired over the same items). The paired difference between the closest open system and Jev is -14.4 points on Belebele (-16.8 to -11.9) and

Model	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.918	0.897 (0.725)	0.751 (0.572)	0.757 (0.481)	0.623 (0.527)
Gemma-4-E4B	0.773	0.853 (0.730)	0.631 (0.519)	0.689 (0.429)	0.590 (0.533)
d3-mini	0.727	0.838 (0.549)	0.729 (0.562)	0.749 (0.383)	0.579 (0.486)
d3-nano	0.702	0.848 (0.377)	0.746 (0.497)	0.705 (0.221)	0.592 (0.408)
Kev-4B	0.561	0.779 (0.451)	0.624 (0.475)	0.575 (0.226)	0.536 (0.408)
d3-lite	0.577	0.760 (0.265)	0.704 (0.463)	0.624 (0.121)	0.535 (0.343)
laya-bangla	0.313	0.750 (0.250)‡	0.625 (0.435)	0.575 (0.077)‡	0.464 (0.216)
d3-edge	0.409	0.475 (0.235)	0.523 (0.445)	0.243 (0.044)	0.472 (0.366)
Kev-0.8B	0.374	0.632 (0.265)	0.469 (0.413)	0.372 (0.118)	0.359 (0.261)
Laya-multilingual	0.287	0.588 (0.275)	0.635 (0.383)	0.179 (0.045)	0.263 (0.141)
Julia-1	0.287	0.520 (0.284)	0.382 (0.316)	–	0.403 (0.299)
Majority class	–	0.250	0.342	0.071	0.195
Uniform random	0.250	0.143	0.333	0.017	0.328

Table 3: Accuracy on Bengali script, and in brackets on rule-based romanization, rotation 0 (SentNoB: macro-F1). Belebele passages were not romanized. Julia-1 cannot take the 60 options of MASSIVE. ‡laya-bangla was fine-tuned on these tasks' training splits. Majority class: the share of the most frequent label in the test items (for SentNoB, the macro-F1 of always predicting it); Belebele's answer positions rotate, so it has none. Uniform random: one over the number of options (for SentNoB, the expected macro-F1 of random guessing). Intervals are in Table 8.

-2.0 on SentNoB (-3.5 to -0.4), both Gemma-4-E4B, and -4.4 on SIB-200 (-8.6 to +0.0), where the interval reaches zero. On XNLI-bn the closest system, d3-nano, is at -0.5 (-2.6 to +1.9), and on MASSIVE d3-mini is at -0.8 (-3.0 to +1.4): there I cannot separate Jev from the best open systems. On its subset, d3-flash is above Jev on XNLI-bn (+7.5 points, +3.3 to +11.9), not distinguishable on MASSIVE, SIB-200 and SentNoB, and below on Belebele (-7.0); the subsets hold only 200 to 405 items per task. Kev-4B lands next to d3-lite: 0.561 on Belebele against 0.577, 0.779 on SIB-200 against 0.760, 0.536 on SentNoB against 0.535, and below it on XNLI-bn (0.624 against 0.704) and MASSIVE (0.575 against 0.624). Kev-0.8B scores 0.632 on SIB-200 and 0.372 on MASSIVE, and falls in the lower group with d3-edge and Laya-multilingual. Both Kev models are well below Jev on every task. Gemma-4-E4B holds up best under rule-based romanization: on SIB-200 it keeps 0.730 of 0.853, where d3-mini keeps 0.549 of 0.838. laya-bangla beats its base model on SIB-200 (0.750 against 0.588) and on MASSIVE (0.575 against 0.179) and still sits below Jev, d3-mini and Gemma-4-E4B on every task. On SentNoB, always answering with the majority class scores a macro-F1 of 0.195 and uniform random guessing 0.328. Laya-multilingual reaches only 0.263, below random guessing, Kev-0.8B 0.359 and Julia-1 0.403. Julia-1 cannot take the 60-option MASSIVE menu at all. Laya-multilingual's SentNoB score also depends on the wording of the question: macro-F1 is 0.263 with my default instruction, 0.321 with option names only, and 0.376 with Bengali wording. I report the default and flag the range.

4.2 Romanization written by native speakers

Table 6 and Figure 2 give the agreement for all twelve systems. Under native-speaker romanization every system kept its Bengali-script answer more often than under rule-based romanization, by +3.3 to +12.9 points, and the interval excludes zero for all 12. For Jev the gain is +6.1 points (95% CI +3.9 to +8.5): it kept its answer on 0.733 of the sentences against 0.672, where chance is about 0.191. The smallest gain is Julia-1's, at +3.3 points, which fits a system that sits near its floor; the largest is d3-edge's, at +12.9. Natural-style romanization, which looks each word up in a corpus

Task	Closest open system	System − Jev, points (95% CI)	d3-flash − Jev on its subset (items)
Belebele	Gemma-4-E4B	-14.4 [-16.8, -11.9]	-7.0 [-10.9, -3.4] (200)
SIB-200	Gemma-4-E4B	-4.4 [-8.6, +0.0]	-0.5 [-4.9, +3.9] (204)
XNLI-bn	d3-nano	-0.5 [-2.6, +1.9]	+7.5 [+3.3, +11.9] (400)
MASSIVE	d3-mini	-0.8 [-3.0, +1.4]	+2.0 [-1.5, +5.0] (403)
SentNoB	Gemma-4-E4B	-2.0 [-3.5, -0.4]	-2.0 [-6.2, +2.0] (405)

Table 4: Paired differences in accuracy (SentNoB: macro-F1) between each task's closest full-set open system and Jev, and between d3-flash and Jev on d3-flash's subset, with 95% cluster-bootstrap intervals. Negative means below Jev.

On the d3-flash subset	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.840	0.897 (0.725)	0.752 (0.557)	0.799 (0.519)	0.637 (0.577)
d3-flash (4-bit)	0.730	0.892 (0.627)	0.828 (0.598)	0.819 (0.412)	0.653 (0.519)

Table 5: d3-flash (4-bit) and Jev on the same hash-ordered subset (204 to 405 items per task); romanized in brackets. d3-flash is the 9B checkpoint in 4-bit quantization.

of typed Banglish, ends up close to the rule-based text. It differs from the rule-based spelling by -1.8 to +4.5 points: it keeps the Bengali answer more often for 7 systems, makes no detectable difference for 5, and is worse for 0. For Jev the difference is -1.8 points (-3.7 to +0.1). Native-speaker romanization keeps the Bengali answer more often than natural-style for all 12 systems, by +1.1 to +8.9 points, and clearly so for 11 of them; for the remaining system the interval includes zero. Tuning a romanizer on typed Banglish therefore gives little, and the gap to what native speakers write remains. The probability vectors tell the same story: the total-variation distance from the Bengali answer is smaller under native-speaker romanization for every system (Table 16).

The ordering of systems moves little. Over the 11 systems with all 1,500 sentences, Kendall τ between the ordering under native-speaker romanization and under rule-based romanization is 0.89. Rule-based romanization therefore ranks systems in nearly the same order as real typing, and overstates how far their answers move. Two limits matter. These are Wikipedia sentences and one topic question, not the five tasks, and the measure is consistency with the system's own Bengali answer, not correctness.

4.3 Calibration and selective prediction

This section covers the 11 systems with full test sets; d3-flash ran on subsets and is left out. Kev has raw ECE, Brier score and confidence gap (Tables 10 and 13) but no development split, so it has no temperature-scaled results; the scaled results cover the other 9 systems that have a development run. Averaged over the three tasks, all 11 systems have confidence above accuracy, but that is not true on every task: 3 of 32 model-task cells have confidence below accuracy (Kev-0.8B, Kev-4B and laya-bangla). Overconfidence differs a lot between models. Raw ECE on Bengali script, averaged over the three tasks as described above, runs from 0.074 for d3-mini to 0.432 for Julia-1 (Table 7). One temperature, fitted on a separate Bengali development split, lowers ECE on Bengali text in all 26 model-task cells where it was fitted; for Julia-1 on XNLI-bn it takes ECE from 0.420 to 0.098. On rule-based romanized text the same temperature leaves a higher ECE for eight of the nine models with a development run. d3-mini is the exception. Jev's scaled ECE is 0.095 on romanized text against 0.047 on Bengali, and laya-bangla's is 0.172 against 0.034. Brier scores are in Table 11; on Bengali script 8 of 32 model-task cells score worse than a uniform guess, and on rule-based

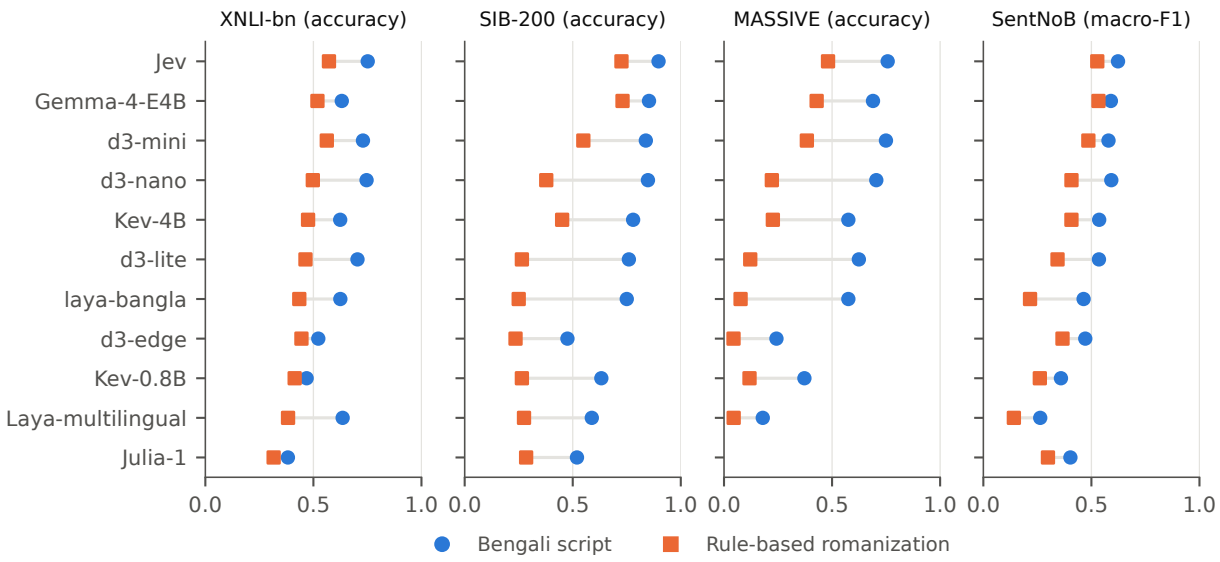


Figure 1: Bengali script against rule-based romanization for each model and task. Lines join the two conditions for the same model.

romanized text 19 of 32. Figure 3 shows the reliability curves and Figure 4 the risk–coverage curves. Confidence still ranks answers within each condition, but rule-based romanization lowers accuracy at every coverage level, including among the most confident answers.

A confidence gate set on Bengali development text to reach 90% accuracy does not carry over. On MASSIVE, Jev accepts 69% of Bengali items (1,036 items) at 91% accuracy, but only 23% of rule-based romanized items (341), at 88%. d3-mini accepts 68% (1,020) at 90% and 22% (335) at 83%. For laya-bangla the accepted rule-based romanized items are 17% accurate, but only 12 items are accepted, so that figure is close to anecdote. So for Jev and d3-mini the gate keeps its accuracy by refusing most rule-based romanized input, and for laya-bangla it fails. This rests on one task and, for some models, on few accepted items, so I treat it as exploratory.

4.4 Option-order sensitivity

Across three rotations of the options, Jev gives the same answer 93% of the time, d3-nano 89%, d3-lite 87% and Gemma-4-E4B 81%. The encoders are far less stable: laya-bangla 67%, Laya-multilingual 53%, Julia-1 35% (Figure 5; per task in Table 9).

4.5 Checks on the harness

Before trusting any of this I tried to reproduce other people's numbers. These checks compare a handful of published numbers; they are not an independent replication of this study. From the Jev authors' released responses, Belebele ben_Beng accuracy is 0.9167; my own run gives 0.918. For Laya-multilingual I get MASSIVE 0.179 and SIB-200 0.588, against 0.18 and 0.57 to 0.58 reported by the laya-bangla authors. For laya-bangla I get SIB-200 0.750 against their 0.76. Resending 300 Jev requests gave the same answer 99% of the time, and a newer Laya package version changed no answer on 6,563 requests. One comparison I could not make: laya-bangla's authors report MASSIVE 0.81 with a two-step scenario-then-intent protocol, and I ask one 60-way question and get 0.575. The study used 99,828 Jev requests, about 53.6M input tokens and $2.25 at $0.042 per

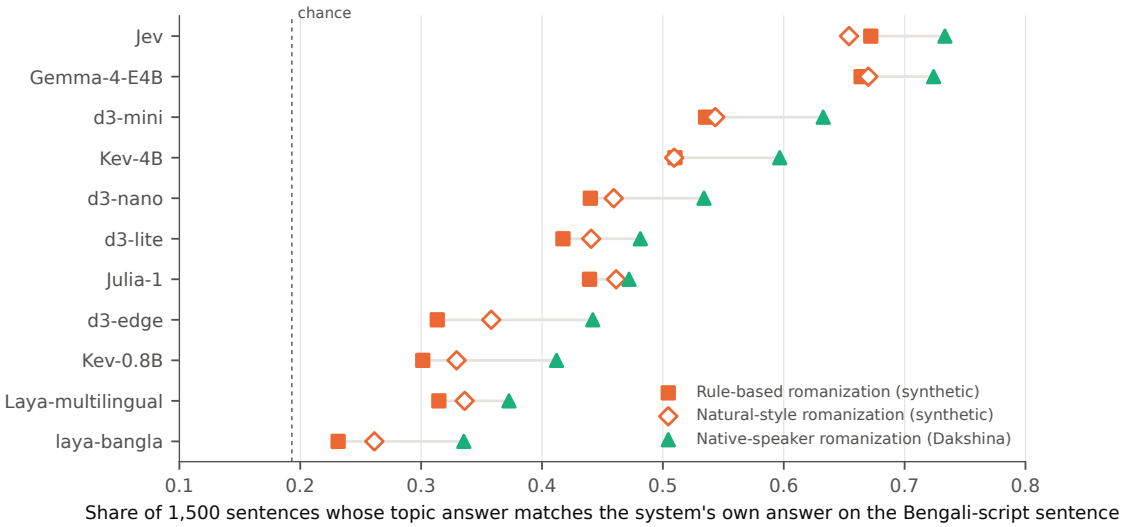


Figure 2: Share of 1,500 Dakshina sentences whose topic answer matches the system's own answer on the Bengali-script sentence, for three spellings of the same sentence. Chance is the dashed line. Intervals are in Table 6; d3-flash (500 sentences) is omitted here.

		Agreement with own Bengali answer			Difference in points (95% CI)	
Model	Sentences	Native-speaker	Natural-style	Rule-based	Native-speaker − rule-based	Native-speaker − natural-style
Jev	1,500	0.733	0.654	0.672	+6.1 [+3.9, +8.5]	+7.9 [+5.6, +10.1]
d3-flash (4-bit)†	500	0.632	0.548	0.558	+7.4 [+3.0, +11.4]	+8.4 [+4.4, +12.6]
Gemma-4-E4B	1,500	0.724	0.670	0.664	+6.0 [+3.8, +8.1]	+5.4 [+3.3, +7.4]
d3-mini	1,500	0.633	0.543	0.535	+9.7 [+7.3, +12.1]	+8.9 [+6.6, +11.3]
d3-nano	1,500	0.534	0.459	0.440	+9.4 [+6.9, +12.0]	+7.5 [+5.1, +9.9]
Kev-4B	1,500	0.597	0.509	0.510	+8.7 [+6.3, +10.9]	+8.7 [+6.3, +11.1]
d3-lite	1,500	0.481	0.441	0.417	+6.4 [+4.0, +8.9]	+4.1 [+1.8, +6.5]
laya-bangla	1,500	0.335	0.261	0.231	+10.4 [+7.5, +13.0]	+7.4 [+4.9, +10.1]
d3-edge	1,500	0.442	0.358	0.313	+12.9 [+10.5, +15.3]	+8.4 [+6.0, +10.9]
Kev-0.8B	1,500	0.412	0.329	0.301	+11.1 [+8.6, +13.5]	+8.3 [+5.7, +10.7]
Laya-multilingual	1,500	0.373	0.336	0.315	+5.8 [+3.3, +8.4]	+3.7 [+0.9, +6.3]
Julia-1	1,500	0.472	0.461	0.439	+3.3 [+0.9, +5.7]	+1.1 [-1.1, +3.2]

Table 6: Agreement with the system's own Bengali-script answer on Dakshina sentences, and paired differences in points with 95% bootstrap intervals. †d3-flash is 4-bit and covers a 500-sentence subset.

million, plus about 30 GPU hours on Kaggle. I measured latency under mixed conditions, so it is not comparable across models, and I do not report it.

5 Discussion

What rule-based romanization can and cannot tell you. Rule-based romanization ranked the systems in nearly the same order as native-speaker romanization, and moved their answers further. For Jev, the rule-based text cost 18.1 points of accuracy on average over four tasks, and the same system kept its answer on 0.672 of the Dakshina sentences under rule-based romanization against 0.733 under native-speaker romanization. So for the systems and sentences I tested, tool-made text moves a system's answers further than native speakers' own spelling does. I did not measure whether it also overstates the accuracy loss on labelled tasks, which would need labelled sentences typed by native speakers. I do not know why the native speakers' spelling is easier. A

	Bengali script		Rule-based romanization	
Model	ECE raw	ECE scaled	ECE raw	ECE scaled
Jev	0.126	0.047	0.146	0.095
Gemma-4-E4B	0.303	0.073	0.391	0.082
d3-mini	0.074	0.036	0.079	0.035
d3-nano	0.083	0.041	0.122	0.054
d3-lite	0.076	0.040	0.172	0.108
laya-bangla	0.204	0.034	0.331	0.172
d3-edge	0.174	0.044	0.202	0.085
Laya-multilingual	0.347	0.080	0.530	0.209
Julia-1	0.432	0.104	0.494	0.154

Table 7: ECE (15 bins) on Bengali script and rule-based romanization, before and after one temperature fitted on a native-text development split. Mean over XNLI-bn, MASSIVE and SentNoB.

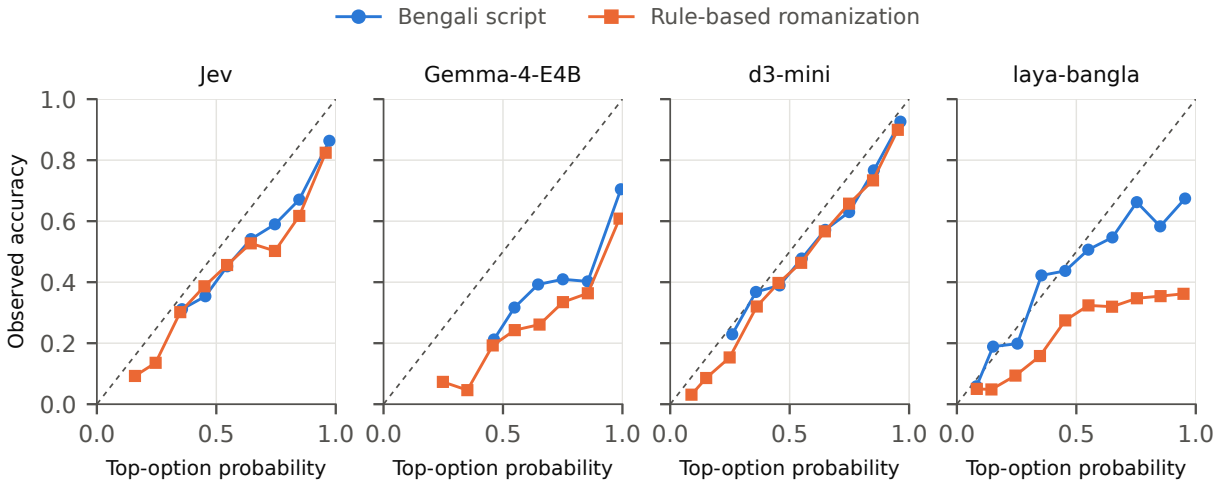


Figure 3: Reliability of four models on MASSIVE, XNLI-bn and SentNoB pooled. The dashed line is perfect calibration.

plausible reason is that people spell common words, and English loan words, the way models have seen them in web text; I did not test that.

Choosing a model. On Bengali script d3-mini, d3-nano and Gemma-4-E4B come within a few points of Jev on several tasks, so the hosted model's advantage here is modest. Gemma-4-E4B is the most robust open model to rule-based romanization, and also among the worst calibrated before scaling (ECE 0.303), which temperature scaling mostly removes. The encoders need care. Their answers change with option order, and Laya-multilingual's SentNoB score moved by more than ten points with the wording of the question.

6 Limitations

The Dakshina comparison measures agreement with a system's own Bengali answer, not accuracy, because the sentences have no labels; a system that is wrong on the Bengali sentence can still agree with itself. It uses Wikipedia sentences and a single seven-way topic question, so it does not show

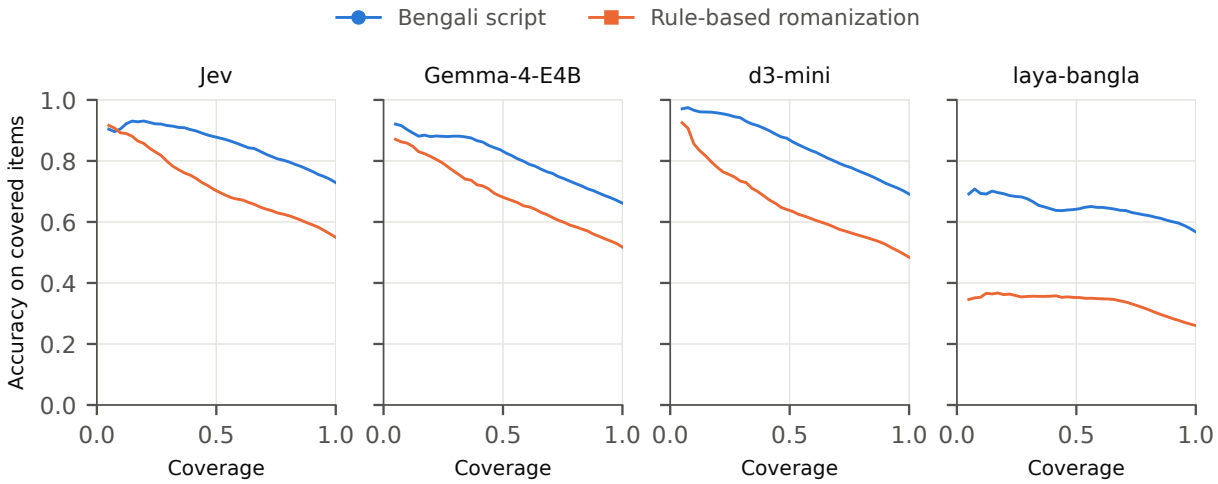


Figure 4: Accuracy on the covered items as coverage grows from the most confident answers.

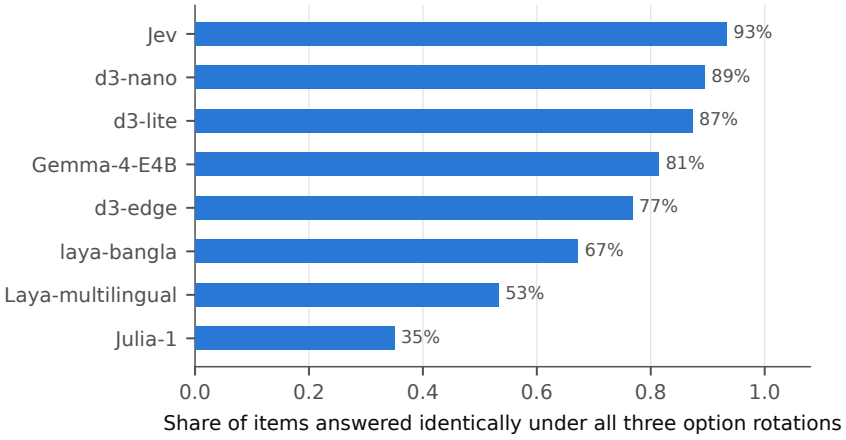


Figure 5: Share of items answered identically under all three option rotations, over all tasks.

that the same gap holds on the five tasks. The sentences were romanized by native speakers, but I do not know how many, or how representative they are of everyday typing on phones and social media. The five tasks use rule-based romanization only. XNLI-bn and MASSIVE are translations, so SentNoB is the only native Bengali text. d3-flash ran 4-bit on subsets, Kev ran only on the test sets and on Dakshina, the larger Kev and d3 models were not run, and Julia-1 cannot take MASSIVE. Contamination is unknown for Jev, Gemma and the Qwen-based d3 models. laya-bangla was fine-tuned on the training splits of MASSIVE and SIB-200 according to its authors, so its scores on those two tasks (marked ‡ in Table 3) are in-distribution and not comparable with the other systems; I could not check overlap with other test items. SentNoB's test split comes from a third-party mirror, differs in size from the other sources (1,586 against 1,567 and 1,564) and overlaps its training split (Section 3). I use one prompt format per model family, and Laya's wording sensitivity shows that this can move a result. The calibration and gate results are exploratory and use no multiplicity correction.

7 Conclusion

On five Bengali tasks Jev leads, several small open models come close on Bengali script, and the open models' overconfidence is largely fixed by one temperature on Bengali text but less fully on rule-based romanized text. Rule-based romanization lowers every system's accuracy, but it moves their answers further than native speakers' own spelling does: on Bengali sentences that native speakers romanized, all twelve systems kept their Bengali-script answer more often than under the rule-based text, while ranking in nearly the same order. Measuring the effect on labelled tasks needs Banglish typed by many writers for sentences that have labels.

Availability. The repository is at `https://github.com/ihpolash/system-one-testing-for-bengali`. It holds the harness, the analysis code that generates every table and number in this paper, the summary result tables, item-level predictions for the open models and per-sentence answers for the Dakshina experiment, SHA-256 checksums for the released files, the exact model identifiers and settings (`docs/MODELS.md`) and the question structure for each task (`docs/PROMPTS.md`). The test-set predictions contain no dataset text except Belebele's option labels, which are the Bengali option texts (CC BY-SA 4.0). The Dakshina predictions are keyed by sentence identifier; `data/make_dakshina.py` rebuilds the sentences from the public dataset (CC BY-SA 4.0). The repository does not contain SentNoB's text: its authors' repository states no licence and I could not confirm its terms. The released test-set predictions do list SentNoB item identifiers, labels and predictions. Licences of all data sources are in Table 14. Jev's raw responses and probability vectors are not included (provider terms); only its top choice and confidence are released for the Dakshina sentences.

Declaration on the use of generative AI

The author used an AI assistant (Claude, Anthropic) for implementation support, analysis and figure scripting, and for drafting and editing this manuscript. All of that work was carried out under the author's direction and review. No generative AI was used to produce any evaluation text: the Latin-script text comes from the Dakshina dataset and from rule-based and corpus-based romanizers. The research questions, benchmark design, evaluation protocol and the interpretation of the results are the author's. All numerical results in the tables and text are generated by the released analysis code from the stored results, and every reference was verified by the author against its primary source. The author takes full responsibility for all content of this paper, irrespective of how it was drafted.

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A Additional tables

Model	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.918 [0.897, 0.936]	0.897 [0.853, 0.936]	0.751 [0.729, 0.772]	0.757 [0.735, 0.779]	0.623 [0.596, 0.647]
Gemma-4-E4B	0.773 [0.743, 0.801]	0.853 [0.804, 0.897]	0.631 [0.606, 0.655]	0.689 [0.667, 0.713]	0.590 [0.566, 0.615]
d3-mini	0.727 [0.697, 0.758]	0.838 [0.789, 0.887]	0.729 [0.705, 0.752]	0.749 [0.728, 0.771]	0.579 [0.555, 0.604]
d3-nano	0.702 [0.671, 0.733]	0.848 [0.799, 0.897]	0.746 [0.725, 0.768]	0.705 [0.681, 0.730]	0.592 [0.568, 0.614]
Kev-4B	0.561 [0.528, 0.592]	0.779 [0.721, 0.833]	0.624 [0.603, 0.648]	0.575 [0.551, 0.600]	0.536 [0.511, 0.556]
d3-lite	0.577 [0.542, 0.609]	0.760 [0.701, 0.819]	0.704 [0.681, 0.726]	0.624 [0.600, 0.647]	0.535 [0.511, 0.558]
laya-bangla	0.313 [0.283, 0.344]	0.750 [0.686, 0.814]	0.625 [0.599, 0.650]	0.575 [0.547, 0.602]	0.464 [0.437, 0.488]
d3-edge	0.409 [0.377, 0.441]	0.475 [0.407, 0.544]	0.523 [0.497, 0.548]	0.243 [0.223, 0.264]	0.472 [0.451, 0.498]
Kev-0.8B	0.374 [0.341, 0.407]	0.632 [0.564, 0.701]	0.469 [0.445, 0.491]	0.372 [0.347, 0.397]	0.359 [0.337, 0.380]
Laya-multilingual	0.287 [0.259, 0.314]	0.588 [0.515, 0.657]	0.635 [0.612, 0.660]	0.179 [0.161, 0.199]	0.263 [0.244, 0.283]
Julia-1	0.287 [0.258, 0.316]	0.520 [0.456, 0.588]	0.382 [0.359, 0.406]	–	0.403 [0.379, 0.424]

Table 8: Accuracy on Bengali script with 95% cluster-bootstrap intervals (SentNoB: macro-F1).

Model	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.9 / 93%	1.0 / 97%	0.5 / 97%	1.7 / 89%	0.9 / 95%
Gemma-4-E4B	1.4 / 74%	0.5 / 93%	3.3 / 86%	4.3 / 70%	0.1 / 94%
d3-nano	1.1 / 82%	0.5 / 96%	0.7 / 96%	1.1 / 85%	0.4 / 95%
d3-lite	1.2 / 82%	1.0 / 96%	0.7 / 96%	0.5 / 80%	1.2 / 92%
laya-bangla	2.1 / 28%	2.0 / 94%	1.2 / 89%	0.3 / 66%	2.8 / 88%
d3-edge	0.3 / 75%	4.4 / 83%	1.9 / 92%	3.3 / 49%	1.4 / 90%
Laya-multilingual	1.1 / 15%	1.0 / 84%	0.1 / 91%	8.7 / 29%	4.6 / 80%
Julia-1	1.8 / 31%	2.9 / 74%	1.9 / 24%	–	10.8 / 45%

Table 9: Spread of accuracy across the three option rotations (points) / share of items answered identically under all three.

	XNLI-bn		MASSIVE		SentNoB	
Model	Bengali	Rule	Bengali	Rule	Bengali	Rule
Jev	0.088	0.197	0.087	0.081	0.202	0.161
Gemma-4-E4B	0.327	0.402	0.249	0.397	0.333	0.374
d3-mini	0.049	0.038	0.054	0.104	0.120	0.094
d3-nano	0.069	0.071	0.073	0.129	0.107	0.165
Kev-4B	0.238	0.316	0.270	0.101	0.170	0.205
d3-lite	0.059	0.057	0.048	0.272	0.120	0.187
laya-bangla	0.203	0.286	0.071	0.138	0.339	0.568
d3-edge	0.178	0.066	0.223	0.423	0.120	0.117
Kev-0.8B	0.173	0.131	0.166	0.053	0.192	0.247
Laya-multilingual	0.276	0.469	0.236	0.468	0.530	0.654
Julia-1	0.420	0.532	–	–	0.444	0.456

Table 10: ECE (15 bins, no temperature scaling) within each task on the full test sets, for Bengali script and rule-based romanization. The mean of these per-task values over the three tasks is the primary ECE of Table 7. d3-flash is omitted (subset).

	XNLI-bn		MASSIVE		SentNoB	
Model	Bengali	Rule	Bengali	Rule	Bengali	Rule
Jev	0.371	0.599	0.341	0.664	0.542	0.590
Gemma-4-E4B	0.684	0.854	0.546	0.919	0.680	0.774
d3-mini	0.377	0.550	0.367	0.763	0.553	0.634
d3-nano	0.352	0.603	0.407	0.905	0.529	0.685
Kev-4B	0.590	0.774	0.645	0.906	0.612	0.726
d3-lite	0.416	0.628	0.507	1.051	0.600	0.738
laya-bangla	0.576	0.777	0.557	1.006	0.817	1.166
d3-edge	0.634	0.644	0.942	1.216	0.657	0.697
Kev-0.8B	0.649	0.688	0.809	0.959	0.743	0.803
Laya-multilingual	0.609	1.029	0.994	1.311	1.124	1.309
Julia-1	0.957	1.147	–	–	0.987	1.014
Uniform guess	0.667		0.983		0.667	

Table 11: Brier score within each task, for Bengali script and rule-based romanization, no temperature scaling (sum over options, mean over items). The last row is the score of a uniform guess, $1 - 1/K$. d3-flash is omitted (subset).

System	All 1,586 items	Items not in train, validation or repeated	Change (points)
Jev	0.623	0.605	-1.8
Gemma-4-E4B	0.590	0.569	-2.1
d3-mini	0.579	0.576	-0.3
d3-nano	0.592	0.589	-0.3
Kev-4B	0.536	0.538	+0.2
d3-lite	0.535	0.536	+0.1
laya-bangla	0.464	0.466	+0.2
d3-edge	0.472	0.477	+0.6
Kev-0.8B	0.359	0.350	-0.9
Laya-multilingual	0.263	0.264	+0.1
Julia-1	0.403	0.419	+1.6

Table 12: SentNoB macro-F1 on all test comments and on those not found in the mirror's training or validation split and not repeated earlier in the test split. d3-flash is omitted (subset).

	Brier score		Confidence − accuracy	
Model	Bengali	Rule-based	Bengali	Rule-based
Jev	0.418	0.617	+0.122	+0.144
Gemma-4-E4B	0.637	0.849	+0.303	+0.391
d3-mini	0.432	0.649	+0.069	+0.073
d3-nano	0.429	0.731	+0.081	+0.121
Kev-4B	0.616	0.802	+0.045	+0.141
d3-lite	0.508	0.806	+0.073	+0.166
laya-bangla	0.650	0.983	+0.159	+0.331
d3-edge	0.744	0.852	+0.173	+0.201
Kev-0.8B	0.734	0.817	+0.066	+0.136
Laya-multilingual	0.909	1.216	+0.347	+0.530
Julia-1	0.972	1.080	+0.431	+0.494

Table 13: Brier score and confidence minus accuracy on Bengali script and rule-based romanization (test sets, mean over XNLI-bn, MASSIVE and SentNoB, no temperature scaling). Positive values mean overconfident. Brier is summed over options and averaged over items, then averaged over tasks; a uniform guess scores 0.772 on this mean (0.667 for Julia-1, which has no MASSIVE).

Dataset	Source and split	Items	SHA-256 (first 12)	Licence
Belebele	facebook/belebele, ben_Beng, test	900	4f06ba1c0800	CC BY-SA 4.0 (dataset card)
SIB-200	Davlan/sib200, ben_Beng, test	204	ef9c5c03dd17	CC BY-SA 4.0 (dataset card)
MASSIVE bn-BD	SetFit/amazon_massive_intent_bn-BD, test (mirror of MASSIVE bn-BD)	2,974	c4bef04b2c50	CC BY 4.0 (original release; the mirror's card gives none)
XNLI-bn	csebuetnlp/xnli_bn, test	4,895	e1be60d7fc23	CC BY-NC-SA 4.0 (dataset card)
SentNoB	sadi2oo3q/SentNoB, test (third-party mirror; 1,586 items, against 1,567 in BLUB and 1,564 in the laya-bangla card)	1,586	8a49a3b96ed7	none stated by the authors; not confirmed
Dakshina-bn	google-research-datasets/dakshina, bn romanized, dev and test (filtered to 8,244)	8,244	a26f9514c478	CC BY-SA 4.0 (repository)

Table 14: Data sources as frozen for this study: identifier and split, number of items, SHA-256 of the frozen file (`data/split_hashes.json`) and licence as stated at the source. Licences were read from the dataset cards and repositories on 24 October 2026. SentNoB's authors' repository states none.

System	Identifier	Revision or version
Jev	`jev-1.13.0 (TypeSafe API)`	hosted, accessed October 2026
d3-edge	`vllm-sr/d3-edge`	18d61df786
d3-lite	`vllm-sr/d3-lite`	a8d25a91ef
d3-nano	`vllm-sr/d3-nano`	3aeb1c22a5
d3-mini	`vllm-sr/d3-mini`	a475367de0
d3-flash	`vllm-sr/d3-flash`	521525effa; 4-bit NF4
Laya-multilingual	`convaiinnovations/laya-multilingual`	not pinned; package laya 0.4.3
laya-bangla	`algolyzer/laya-bangla`	v2, 4c5eb48932; package laya 0.4.11
Julia-1	`SupersonicLabs/Julia-1`	a85b127321
Kev-0.8B, Kev-4B	`jaredpalmer/kev-0.8b, kev-4b`	tag v1.0; code 2ecafc49fe
Gemma-4-E4B	`google/gemma-4-E4B-it`	not pinned

Table 15: Identifiers and revisions of the systems. Exact settings are in `docs/MODELS.md`.

	Probability shift (TV)			Mean top-option confidence		
Model	Native-speaker	Natural-style	Rule-based	Native-speaker	Rule-based	Bengali
Jev	0.262	0.336	0.324	0.749	0.711	0.848
d3-flash (4-bit)†	0.324	0.399	0.392	0.632	0.597	0.764
Gemma-4-E4B	0.271	0.322	0.325	0.910	0.905	0.937
d3-mini	0.346	0.430	0.439	0.655	0.594	0.792
d3-nano	0.412	0.492	0.506	0.621	0.566	0.773
Kev-4B	0.310	0.382	0.383	0.550	0.498	0.655
d3-lite	0.453	0.506	0.529	0.570	0.511	0.759
laya-bangla	0.566	0.626	0.646	0.589	0.543	0.759
d3-edge	0.452	0.491	0.509	0.545	0.452	0.636
Kev-0.8B	0.412	0.465	0.480	0.515	0.466	0.605
Laya-multilingual	0.608	0.648	0.670	0.850	0.837	0.855
Julia-1	0.528	0.547	0.565	0.806	0.850	0.830

Table 16: Dakshina sentences: mean total-variation distance between the probability vector on the Bengali sentence and on each romanization (0 is identical, 1 disjoint), and mean top-option confidence on each spelling. d3-flash is 4-bit and covers a 500-sentence subset.